Assignment – Solving a Search Problem from Scratch (Up to A*)

Assignment Title

*From Problem Definition to A Search: Building an Intelligent Path-Finding Agent**

Problem Description

You are given a **grid-based map** representing a small city. An AI agent must move from a **start position (S)** to a **goal position (G)** while avoiding obstacles.

```
S  .  .  .
#  #  .  #
.  .  .  G
```

Legend:

- **S** = Start
- **G** = Goal
- **.** = Free cell
- **#** = Obstacle (cannot be crossed)

The agent can move **up, down, left, right**. Each move has a **cost of 1**.

Apply A* Step by Step

1. For each expanded state, compute:
    - g(n)
    - h(n)
    - f(n) = g(n) + h(n)
2. Show which node A* selects at each step
3. Stop when the goal is reached

Expected Outcome

- A*: finds the **shortest path efficiently**

Submission Requirements

- Python implementation of A*